

Paarth Bhasin

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INTRODUCTION

A seasoned **Software Engineer / SRE** with **7+** years of experience, and strong problem-solving skills, specializing in creating bespoke digital products and experiences.

With a rare combination of skills in **Backend & Frontend** development, **Site Reliability Engineering**, and a unique and deep insight into products and customers, that enables me to build world-class Software products.

EXPERIENCE

▷ **Senior Full Stack Engineer (Financial, Rewards & Accounting)** Nov 2024 - Present
Big WX / Woolworths Group (<https://bigw.com.au>) *Sydney, NSW*

Responsibilities:

- E-commerce feature design and development, responsible for the order checkout, rewards/discounts, campaigns and loyalty card features on the website as well as the back-end. This serves traffic in the scale of 1M+ members weekly.
- Collaborate with Platform team to implement dedicated **Kubernetes** pods setup for the **Next.js** app - APIs and UI.
- Enhance observability and monitoring tooling, at production level for the front-end by utilizing **OpenTelemetry** coupled with Sentry to capture relevant traces.
- Working with **TypeScript**, **Kubernetes**, **GraphQL**, **Next.js**, **Material UI / Shadcn**, **Node.js**, **C#**, **Java**, **React-native/Swift**, **CSS/SCSS** & **Azure** to develop features related to digital customer rewards / loyalty cards. This involves improving business reporting requirements (financial reconciliation and accounting), critical bug-fixes as well as UX/UI improvements, and end to end testing improvements.

Achievements:

- Increased cart checkout rates from 56% to 70% for customers with a loyalty card.
- Refactored front-end to send order tracking analytics to critical services like Google Cloud Analytics, more reliably by removing dependency on downstream services and removing inefficiencies like duplicates.
- Led the development of **PayPal** and **AfterPay** payment methods on checkout, and a dynamically responsive Order checkout page, responsible for \$15M+ in sales.
- Increased end to end test coverage of critical user walkthrough scenarios (eg. cart page, checkout etc.) in front-end, following TDD.

▷ **Full Stack Engineer** Jun 2024 - Nov 2024
Evermed (<https://evermed.com.au>) *Sydney, NSW*

Responsibilities:

- Part of a 4 person startup, empowering customers across Australia, effectively acting in the capacity of tech lead.
- Rollout of new e-commerce web & back-end features, optimizing existing features, reducing tech-debt and improving operations in a fast-paced environment., with lots of pair-programming.
- Working with **AWS**, **Python**, **C#/.NET**, **CSS**, **Node.js**, **React.js**, **Material UI**, **Nginx** for full-stack application development. Introduced test based approach cloud-infra and front-end applications, outlining testing strategy. Also performed user experience research.

Achievements:

- Significantly improved the frontend admin portal for pharmacy staff, to produce much more detailed information about customers and prescriptions, using CSS best practices.
- Presented an alternative to subsection of internal micro-services design to instead use **GraphQL**.

▷ Research Engineer - Front End / AI

MapAI (<https://mapai.net>)

Apr 2024 - Jun 2024

Sydney, NSW

Responsibilities:

- Project funded by New South Wales Government, working as part of a university initiated research project.
- Full stack application design with best-practices for geo-spatial analysis tools working with Data Scientists.
- This also involved running the machine learning model on a personalized high powered GPU machine provisioned by the university, which was configured to be accessible by the Internet through AWS VPC public IP.

Achievements:

- Built a hoverable chat experience, that could run queries and keep history.
- Contributed to the LLM model integration with OpenAI, using **Python LangChain**.

▷ Senior Software Engineer - Platform/SRE/Performance

Atlassian

Jan 2019 - Mar 2024

Sydney, NSW

Responsibilities:

- Make Jira reliable by introducing key improvements in the application (**cache, DB optimization, algorithmic improvements, traffic proxying, rate-limiting, OS settings**), whilst maintaining 99.9% uptime SLA.
- Participated in regular on-call roster for troubleshooting and incident management for Jira in AWS cloud.
- Led project to define ownership of monitoring and key operational knowledge elements within the team, to have it fully functional and ready to operate as a dedicated SRE team.
- Managed all aspects of **CI/CD** pipelines - containerisation, deployments, and configuration as code in **Terraform**. This involved large scale scripting in **Terraform**, using best-practices like modules, remote state management, variables, version-management (**tfenv**), loops/counters.

Key Achievements:

- Led initiative for performance optimization of a widely used user-facing endpoint serving millions of customers, by *algorithm redesign* & **SQL** query optimisation, improving endpoint latency by 50% on average, eg **91s → 35s**. This involved lots of collaboration and pair-programming across different geographies.
- Developed a UI, used for monitoring, tracing & management of deployments for synthetic tests, using **GraphQL** (using **Graphene** based **Python GraphQL** server), **React Relay** & **TypeScript**. It is used by 50+ engineering teams (>1000 developers) daily.
- Drove **OpenTelemetry** adoption in Jira in **Java**, to adopt observability best-practices and for improving the tracing footprint. This involved evaluation of third-party tools as well such as **Grafana**, **NewRelic**, **Honeycomb.io**, **Prometheus**. This also involved *open source* contributions to OpenTelemetry's Java framework.
- Mentored junior engineers, to improve their working dynamics in a globally distributed team environment.
- Led an initiative to improve resiliency of database towards cluster fail-over, requiring OS level programming related with **JDBC** database driver and **TCP/IP** layer network optimization to reduce time to recover (TTR), for (**Aurora**) across the fleet in multiple regions/data-centers. This reduced TTR from 30-60m to 45-60s on average and <1s for best case scenarios.
- Devised testing framework for **performance analysis** of large customers in production coming to cloud, for multiple feature teams, playing key role in unblocking the migration of large customers (like Equifax - 50k+ users) to adopt our cloud services. This developed an underp-pinning strategy and principles for reliability, performance and scale testing strategy to conduct such testing for entire Jira Development organisation, in various geographies (~1500-2000 Developers).
- Developed automation (**Python**) for auto-scaling of **WebServer (AWS EC2)**, to handle to high tomcat thread loads in Jira, improving downtime during peak traffic, and TTR to <5m, TTD < 5s and uptime at 99.9% SLA.

EDUCATION

Masters of Computer Engineering, *University of New South Wales - Distinction, WAM: 71.0*
(Part-time)

2019 - 2022

Bachelor of Software Engineering, *Monash University*

2014 - 2018